Forming Equations from Words

Exam code: 0607

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Forming linear equations

How do I form expressions from words?

- You can turn common **phrases** into expressions
  - **Use *x*** to represent an **unknown** value

2 less than "something"	$x - 2$
Double "something"	$2x$
5 lots of "something"	$5x$
3 more than "something"	$x + 3$
Half of "something"	$\frac{1}{2}x$ or $\frac{x}{2}$

- Common words indicating basic operations are:
  - **Addition**: sum, total, more than, increase
  - **Subtraction**: difference, less than, decrease
  - **Multiplication**: product, lots of, times as many, double, triple
  - **Division**: shared, split, grouped, halved, quartered
- **Brackets** help keep the **order** correct
  - "something" add 1, then multiplied by 3
    - $(x + 1) \times 3$ which simplifies to $3(x + 1)$
  - Compare this to "something" multiplied by 3, then add 1
    - $x \times 3 + 1$ which simplifies to $3x + 1$
- You may have to **choose** which unknown to call *x*
  - If Adam is 10 years **younger** than Barry, then Barry is 10 years **older** than Adam
    - Either represent Adam's age as $x - 10$ and Barry's age as x
    - Or represent Adam's age as x and Barry's age as $x + 10$

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- This makes the algebra easier (rather than using x for Barry's age and $\frac{1}{2}x$ for Adam's age)

How do I form equations?

- An **equation** is a statement with an **equals** sign that can be **solved**
- Try to put in the phrase "**is equal to**" to see where the equals goes
  - Lisa's age is double Aisha's age and the sum of their ages is ("is equal to") 27
    - Represent Aisha's age as x and Lisa's age is $2x$
    - The equation is $2x + x = 27$
  - When solving, always give the answer in context
    - $3x = 27$ so $x = 9$
    - In context: "Lisa is 18 years old and Aisha is 9 years old"
- Sometimes you might have **two** unknown values (x and y)
  - Use the information to form two **simultaneous** equations

Worked Example

A flowerbed has flowers of three different colours: red, yellow and purple.
The number of yellow flowers is three times the number of red flowers.
The number of purple flowers is 5 more than the number of yellow flowers.

If the difference between the number of purple flowers and red flowers is 29, find the number of yellow flowers.

Answer:

Let the number of red flowers be x

$$x \text{ red flowers}$$

Multiply this by 3 to get the number of yellow flowers

$$3x \text{ yellow flowers}$$

Add 5 to the previous result to get the number of purple flowers

$$3x + 5 \text{ purple flowers}$$

Find the difference between the number of purple and red flowers (purple subtract red, as purple is larger)

$$3x + 5 - x$$

Set the difference equal to 29

$$3x + 5 - x = 29$$

Solve the equation (subtract 5 then divide by 2)

$$2x = 29 - 5$$
$$2x = 24$$
$$x = \frac{24}{2}$$
$$x = 12$$

This is not the answer to the question asked
The number of yellow flowers is $3x$ so multiply this answer by 3

There are 36 yellow flowers

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